

HAL CANON

BOOK IV

Component Specifications

A controlled family defining what every HAL subsystem must do, own, expose, protect, prove, recover, and never bypass.

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Book I is supreme. Book II is authoritative. Book III governs engineering. Book X governs shared meaning.

Document control

Document	HAL Book IV - Component Specifications
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Authority	Subordinate to Books I, II, and III; aligned to Book X
Owner Review	No open item

Authority statement

Book IV refines Book II into implementation-ready component obligations. It cannot change Books I-III, redefine Book X, or independently define Book IX wire contracts. Higher-order sources always control.

Collection rules

- Every authoritative state domain has exactly one mutation owner.
- Logical interfaces define semantic obligations; Book IX defines machine contracts.
- No component gains Authority from data possession, code execution, credentials, trust, usefulness, or deployment locality.
- Every critical invariant, authority path, trust boundary, lifecycle transition, failure mode, recovery path, privacy obligation, and prohibited shortcut has a conformance method.
- Component specialization cannot redefine Book X Canonical Terms.

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CMP-01 - Constitutional Kernel

Final v1.0 | Book II Chapters 01, 03, 05, 20, 29, 35 | Component certification: PASS

Purpose and scope

Admit or reject protected actions and protected state transitions while preserving HAL's constitutional identity.

Authority and sources

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 01, 03, 05, 20, 29, 35. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony.

Authoritative state

Constitutional Identity Record; Constitution and Invariant Registry; protected-policy admission metadata; Owner Authorization Ceremony Registry; Protected Audit Anchor; Recovery Admission Record. CMP-01 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

natural-language interpretation; general planning; ordinary policy administration; general evidence custody. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

No protected effect occurs without current identity, authority, policy, evidence, verification, and exact-change checks.

Lifecycle

Normal > Restricted > Safe > Safe Recovery > Normal. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain root-state uncertainty, Owner identity conflict, replayed Authority or Permission decision, audit-anchor mismatch, replica disagreement. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Constitutional Identity Record.
- Own and protect mutation of Constitution and Invariant Registry.
- Own and protect mutation of protected-policy admission metadata.
- Own and protect mutation of Owner Authorization Ceremony Registry.
- Own and protect mutation of Protected Audit Anchor.
- Own and protect mutation of Recovery Admission Record.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-01-IF-01 / Command	Evaluate Protected Action	Command `Evaluate Protected Action` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-02 / Command	Commit Protected Mutation	Command `Commit Protected Mutation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-03 / Command	Validate Owner Authorization	Command `Validate Owner Authorization` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-04 / Command	Admit Constitutional Recovery	Command `Admit Constitutional Recovery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-05 / Query	Get Constitutional State	Query `Get Constitutional State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-06 / Query	Get Protected Decision	Query `Get Protected Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-07 / Event	Protected Action Admitted	Event `Protected Action Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-08 / Event	Protected Action Denied	Event `Protected Action Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-01-IF-09 / Event	Constitutional Mode Changed	Event `Constitutional Mode Changed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-01-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-01-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-01-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-01-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-01-TST-005 - Critical invariant: No protected effect occurs without current identity, authority, policy, evidence, verification, and exact-change checks. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-01-TST-006 - Failure containment: Inject root-state uncertainty, Owner identity conflict, replayed Authority or Permission decision, audit-anchor mismatch, replica disagreement. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-01-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-01-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-01-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-01-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58, Book II Chapters 01, 03, 05, 20, 29, 35, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-02 - Identity and Continuity Service

Final v1.0 | Book II Chapters 04, 28 | Component certification: PASS

Purpose and scope

Maintain durable governed identity and continuity for HAL, humans, services, devices, nodes, and Presences.

Authority and sources

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 04, 28. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Principal, Identity Record, Identifier, Continuity, Authentication.

Authoritative state

Identity Records; identifier namespaces; continuity lineage; identity lifecycle; authentication-assurance bindings. CMP-02 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Owner authority; delegations; trust scores; credentials or secret material. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

One constitutional HAL identity persists across all Presences, runtimes, models, services, and recovery events.

Lifecycle

Proposed > Active > Suspended > Retired; continuity: Unverified > Verified > Disputed. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain duplicate HAL identity, identifier collision, lineage break, stale authentication binding, identity-source disagreement. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Identity Records.
- Own and protect mutation of identifier namespaces.
- Own and protect mutation of continuity lineage.
- Own and protect mutation of identity lifecycle.
- Own and protect mutation of authentication-assurance bindings.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-02-IF-01 / Command	Create Identity	Command 'Create Identity' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-02 / Command	Bind Identifier	Command 'Bind Identifier' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-03 / Command	Suspend Identity	Command 'Suspend Identity' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-04 / Command	Reconcile Continuity	Command 'Reconcile Continuity' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-05 / Query	Resolve Identity	Query 'Resolve Identity' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit

		success or denial semantics. Book IX owns wire details.
CMP-02-IF-06 / Query	Get Continuity Lineage	Query `Get Continuity Lineage` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-07 / Query	Get Authentication Context	Query `Get Authentication Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-08 / Event	Identity Created	Event `Identity Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-09 / Event	Identity Suspended	Event `Identity Suspended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-02-IF-10 / Event	Continuity Reconciled	Event `Continuity Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-02-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-02-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-02-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-02-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-02-TST-005 - Critical invariant: One constitutional HAL identity persists across all Presences, runtimes, models, services, and recovery events. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-02-TST-006 - Failure containment: Inject duplicate HAL identity, identifier collision, lineage break, stale authentication binding, identity-source disagreement. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-02-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-02-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-02-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-02-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 35, 40, 47-50, and 58, Book II Chapters 04, 28, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Principal, Identity Record, Identifier, Continuity, Authentication. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-03 - Authority and Delegation Service

Final v1.0 | Book II Chapters 05 | Component certification: PASS

Purpose and scope

Own bounded authority, delegation, revocation, and policy-decision records without inferring authority from identity, trust, credentials, or capability.

Authority and sources

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 05. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Permission, Delegation, Policy Decision Record.

Authoritative state

Authority Grants; Delegation Records; revocations; Policy Decision Records; authority-context envelopes. CMP-03 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

authoritative Identity state; trust assessment; capability implementation; constitutional root authority. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Every delegation is attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's current authority.

Lifecycle

Draft > Active > Suspended > Revoked or Expired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain delegation cycle, scope escalation, stale revocation, policy-version mismatch, authority-context loss. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Authority Grants.
- Own and protect mutation of Delegation Records.
- Own and protect mutation of revocations.
- Own and protect mutation of Policy Decision Records.
- Own and protect mutation of authority-context envelopes.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-03-IF-01 / Command	Grant Delegation	Command `Grant Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-02 / Command	Revoke Delegation	Command `Revoke Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-03 / Command	Evaluate Permission	Command `Evaluate Permission` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-04 / Command	Expire Delegation	Command `Expire Delegation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-05 / Query	Get Effective Authority	Query `Get Effective Authority` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-06 / Query	Explain Policy Decision	Query `Explain Policy Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-07 / Query	List Active Delegations	Query `List Active Delegations` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-08 / Event	Delegation Activated	Event `Delegation Activated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-09 / Event	Delegation Revoked	Event `Delegation Revoked` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-03-IF-10 / Event	Permission Decided	Event `Permission Decided` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-03-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-03-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-03-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-03-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-03-TST-005 - Critical invariant: Every delegation is attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's current authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-03-TST-006 - Failure containment: Inject delegation cycle, scope escalation, stale revocation, policy-version mismatch, authority-context loss. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-03-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-03-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-03-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-03-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 35, 40, 47-50, and 58, Book II Chapters 05, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Permission, Delegation, Policy Decision Record. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-04 - Intent Service

Final v1.0 | Book II Chapters 06 | Component certification: PASS

Purpose and scope

Capture attributable intent and maintain traceability from Intent through Goals, Objectives, Plans, Tasks, and outcome criteria.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 06. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Intent, Goal, Objective, Plan, Task.

Authoritative state

Intent Records; Goal and Objective hierarchy; intent version lineage; acceptance and outcome criteria. CMP-04 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

plan execution; resource placement; judgment; action Permission decision. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Execution artifacts remain attributable to an authorized Intent and cannot silently broaden its scope.

Lifecycle

Proposed > Clarifying > Accepted > Active > Satisfied, Abandoned, or Superseded. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain ambiguous principal, goal drift, orphan objective, conflicting intent version, untraceable task. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Intent Records.
- Own and protect mutation of Goal and Objective hierarchy.
- Own and protect mutation of intent version lineage.
- Own and protect mutation of acceptance and outcome criteria.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-04-IF-01 / Command	Register Intent	Command 'Register Intent' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-02 / Command	Revise Intent	Command 'Revise Intent' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-03 / Command	Decompose Goal	Command 'Decompose Goal' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-04 / Command	Retire Intent	Command 'Retire Intent' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-05 / Query	Get Intent Graph	Query 'Get Intent Graph' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-06 / Query	Trace Objective	Query 'Trace Objective' MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-07 / Query	List Outcome Criteria	Query `List Outcome Criteria` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-08 / Event	Intent Registered	Event `Intent Registered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-09 / Event	Intent Revised	Event `Intent Revised` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-04-IF-10 / Event	Objective Retired	Event `Objective Retired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-04-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-04-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-04-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-04-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-04-TST-005 - Critical invariant: Execution artifacts remain attributable to an authorized Intent and cannot silently broaden its scope. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-04-TST-006 - Failure containment: Inject ambiguous principal, goal drift, orphan objective, conflicting intent version, untraceable task. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-04-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-04-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-04-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-04-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 06, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Intent, Goal, Objective, Plan, Task. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-05 - Cognitive Orchestrator

Final v1.0 | Book II Chapters 07 | Component certification: PASS

Purpose and scope

Coordinate bounded cognitive work, model and tool routing, execution graphs, and result assembly without gaining authority to act.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 07. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Execution Graph, Provider, Capability.

Authoritative state

Execution Graphs; orchestration runs; provider-selection records; result assembly state. CMP-05 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Intent meaning; attention priority; judgment; capability authority; real-world commit. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Orchestration coordinates work but never converts provider availability, model output, or useful results into authority.

Lifecycle

Queued > Running > Waiting > Completed, Failed, or Cancelled. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain provider hallucination, graph deadlock, unbounded recursion, result provenance loss, cancel failure. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Execution Graphs.
- Own and protect mutation of orchestration runs.
- Own and protect mutation of provider-selection records.
- Own and protect mutation of result assembly state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-05-IF-01 / Command	Start Orchestration	Command `Start Orchestration` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-02 / Command	Advance Execution Graph	Command `Advance Execution Graph` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-03 / Command	Cancel Run	Command `Cancel Run` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-04 / Command	Request Provider Work	Command `Request Provider Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-05 / Query	Get Run State	Query `Get Run State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-06 / Query	Explain Routing	Query `Explain Routing` MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-07 / Query	Get Execution Graph	Query `Get Execution Graph` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-08 / Event	Run Started	Event `Run Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-09 / Event	Node Completed	Event `Node Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-10 / Event	Run Failed	Event `Run Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-05-IF-11 / Event	Result Assembled	Event `Result Assembled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-05-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-05-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-05-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-05-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-05-TST-005 - Critical invariant: Orchestration coordinates work but never converts provider availability, model output, or useful results into authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-05-TST-006 - Failure containment: Inject provider hallucination, graph deadlock, unbounded recursion, result provenance loss, cancel failure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-05-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-05-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-05-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-05-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.

- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 07, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Execution Graph, Provider, Capability. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-06 - Attention Manager

Final v1.0 | Book II Chapters 08 | Component certification: PASS

Purpose and scope

Admit, prioritize, place, and preempt bounded work under declared resource, urgency, fairness, and constitutional constraints.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 08. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Attention Object, Resource, Reservation.

Authoritative state

Attention Objects; work-admission decisions; priority and preemption records; placement leases. CMP-06 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Intent; Judgment; resource inventory; action permission. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

The Attention Manager is the sole authority for work admission and placement; urgency cannot bypass constitutional or authority controls.

Lifecycle

Requested > Admitted > Running > Preempted, Completed, or Rejected. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain priority inversion, starvation, resource overcommit, stale lease, unbounded queue. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Attention Objects.
- Own and protect mutation of work-admission decisions.
- Own and protect mutation of priority and preemption records.
- Own and protect mutation of placement leases.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-06-IF-01 / Command	Submit Attention Request	Command `Submit Attention Request` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-02 / Command	Admit Work	Command `Admit Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-03 / Command	Preempt Work	Command `Preempt Work` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-04 / Command	Release Placement	Command `Release Placement` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-05 / Query	Get Queue State	Query `Get Queue State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-06 / Query	Explain Priority	Query `Explain Priority` MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-07 / Query	Get Placement	Query `Get Placement` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-08 / Event	Work Admitted	Event `Work Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-09 / Event	Work Preempted	Event `Work Preempted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-06-IF-10 / Event	Placement Released	Event `Placement Released` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-06-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-06-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-06-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-06-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-06-TST-005 - Critical invariant: The Attention Manager is the sole authority for work admission and placement; urgency cannot bypass constitutional or authority controls. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-06-TST-006 - Failure containment: Inject priority inversion, starvation, resource overcommit, stale lease, unbounded queue. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-06-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-06-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-06-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-06-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 08, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Attention Object, Resource, Reservation. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-07 - Judgment Service

Final v1.0 | Book II Chapters 09 | Component certification: PASS

Purpose and scope

Produce evidence-linked Decision Objects that expose alternatives, uncertainty, values, authority context, rationale, and review conditions.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 09. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Judgment, Decision Object, Uncertainty.

Authoritative state

Decision Objects; alternative evaluations; uncertainty assessments; review conditions. CMP-07 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Intent; authority grants; evidence custody; action execution. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

A Decision Object records judgment but does not itself create Permission, Authority, or a real-world effect.

Lifecycle

Proposed > Deliberating > Decided > Under Review > Superseded or Closed. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain missing alternative, hidden uncertainty, authority laundering, stale evidence, irreproducible rationale. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Decision Objects.
- Own and protect mutation of alternative evaluations.
- Own and protect mutation of uncertainty assessments.
- Own and protect mutation of review conditions.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-07-IF-01 / Command	Request Judgment	Command `Request Judgment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-02 / Command	Record Decision	Command `Record Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-03 / Command	Reconsider Decision	Command `Reconsider Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-04 / Command	Withdraw Decision	Command `Withdraw Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-05 / Query	Get Decision Object	Query `Get Decision Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-06 / Query	Explain Rationale	Query `Explain Rationale` MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-07 / Query	List Alternatives	Query `List Alternatives` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-08 / Event	Decision Recorded	Event `Decision Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-09 / Event	Decision Reconsidered	Event `Decision Reconsidered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-07-IF-10 / Event	Decision Withdrawn	Event `Decision Withdrawn` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-07-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-07-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-07-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-07-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-07-TST-005 - Critical invariant: A Decision Object records judgment but does not itself create Permission, Authority, or a real-world effect. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-07-TST-006 - Failure containment: Inject missing alternative, hidden uncertainty, authority laundering, stale evidence, irreproducible rationale. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-07-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-07-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-07-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-07-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 09, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Judgment, Decision Object, Uncertainty. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-08 - Knowledge Service

Final v1.0 | Book II Chapters 10 | Component certification: PASS

Purpose and scope

Maintain provenance-aware knowledge representations and retrieval while distinguishing claims, evidence, confidence, validity, and authoritative state.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 10. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Knowledge, Knowledge Graph, Claim, Provenance.

Authoritative state

Knowledge Graph; knowledge assertions; provenance links; validity and confidence metadata; retrieval indexes. CMP-08 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Evidence Objects; Experience Ledger; Memory Graph; authoritative operational state. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

No assertion becomes Knowledge without provenance, domain, time, confidence, and challenge semantics.

Lifecycle

Candidate > Admitted > Qualified > Challenged > Superseded or Retired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain provenance loss, stale assertion, index-authority confusion, causation inference, unbounded disclosure. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Knowledge Graph.
- Own and protect mutation of knowledge assertions.
- Own and protect mutation of provenance links.
- Own and protect mutation of validity and confidence metadata.
- Own and protect mutation of retrieval indexes.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-08-IF-01 / Command	Admit Knowledge Assertion	Command `Admit Knowledge Assertion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-02 / Command	Revise Knowledge Link	Command `Revise Knowledge Link` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-03 / Command	Invalidate Assertion	Command `Invalidate Assertion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-04 / Command	Rebuild Index	Command `Rebuild Index` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-05 / Query	Query Knowledge	Query `Query Knowledge` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-06 / Query	Trace Provenance	Query `Trace Provenance` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-07 / Query	Get Validity Context	Query `Get Validity Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-08 / Event	Knowledge Admitted	Event `Knowledge Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-09 / Event	Assertion Invalidated	Event `Assertion Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-08-IF-10 / Event	Index Rebuilt	Event `Index Rebuilt` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-08-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-08-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-08-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-08-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-08-TST-005 - Critical invariant: No assertion becomes Knowledge without provenance, domain, time, confidence, and challenge semantics. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-08-TST-006 - Failure containment: Inject provenance loss, stale assertion, index-authority confusion, causation inference, unbounded disclosure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-08-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-08-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-08-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-08-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 10, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Knowledge, Knowledge Graph, Claim, Provenance. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-09 - Memory System

Final v1.0 | Book II Chapters 12 | Component certification: PASS

Purpose and scope

Maintain contextual associations and governed retrieval of retained experience without confusing memory, knowledge, cache, or authoritative state.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 12. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Memory, Memory Graph, Experience.

Authoritative state

Memory Graph; memory associations; retention and access metadata; Conversation Objects where assigned. CMP-09 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Knowledge Graph; Evidence Objects; Experience Ledger custody; identity or authority. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Memory is provenance-linked, purpose-bound, and never silently promoted to knowledge or evidence.

Lifecycle

Candidate > Active > Restricted > Expired or Deleted. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain cross-principal leakage, retention violation, false association, conversation split, deletion failure. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Memory Graph.
- Own and protect mutation of memory associations.
- Own and protect mutation of retention and access metadata.
- Own and protect mutation of Conversation Objects where assigned.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-09-IF-01 / Command	Store Memory Association	Command `Store Memory Association` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-02 / Command	Revise Association	Command `Revise Association` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-03 / Command	Forget Memory	Command `Forget Memory` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-04 / Command	Reconcile Conversation	Command `Reconcile Conversation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-05 / Query	Recall Memory	Query `Recall Memory` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-06 / Query	Trace Memory Source	Query `Trace Memory Source` MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-07 / Query	Get Retention Status	Query `Get Retention Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-08 / Event	Memory Associated	Event `Memory Associated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-09 / Event	Memory Forgotten	Event `Memory Forgotten` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-09-IF-10 / Event	Conversation Reconciled	Event `Conversation Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-09-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-09-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-09-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-09-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-09-TST-005 - Critical invariant: Memory is provenance-linked, purpose-bound, and never silently promoted to knowledge or evidence. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-09-TST-006 - Failure containment: Inject cross-principal leakage, retention violation, false association, conversation split, deletion failure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-09-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-09-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-09-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-09-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 12, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Memory, Memory Graph, Experience. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-10 - Learning and Wisdom System

Final v1.0 | Book II Chapters 11, 12 | Component certification: PASS

Purpose and scope

Convert governed experience into reproducible patterns and bounded wisdom while preventing silent self-modification of protected behavior.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 11, 12. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Learning, Pattern, Wisdom, Experience Ledger.

Authoritative state

Experience Ledger; Pattern Records; Wisdom Records; learning-run records; promotion decisions. CMP-10 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

constitutional meaning; authority; deployment into a declared live-effect environment and approved Reality Boundary stage; Knowledge Graph. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Learning cannot silently modify constitutional meaning, authority, protected behavior, or authoritative live-effect state.

Lifecycle

Experience: Recorded > Validated > Retained; Pattern: Candidate > Qualified > Approved > Retired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain training-data poisoning, pattern overgeneralization, silent promotion, feedback loop, unbounded self-modification. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Experience Ledger.
- Own and protect mutation of Pattern Records.
- Own and protect mutation of Wisdom Records.
- Own and protect mutation of learning-run records.
- Own and protect mutation of promotion decisions.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-10-IF-01 / Command	Record Experience	Command `Record Experience` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-02 / Command	Propose Pattern	Command `Propose Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-03 / Command	Promote Pattern	Command `Promote Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-04 / Command	Retire Wisdom	Command `Retire Wisdom` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-05 / Query	Get Experience	Query `Get Experience` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit

		success or denial semantics. Book IX owns wire details.
CMP-10-IF-06 / Query	Explain Pattern	Query `Explain Pattern` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-07 / Query	Get Wisdom Limits	Query `Get Wisdom Limits` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-08 / Event	Experience Recorded	Event `Experience Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-09 / Event	Pattern Promoted	Event `Pattern Promoted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-10-IF-10 / Event	Wisdom Retired	Event `Wisdom Retired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-10-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-10-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-10-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-10-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-10-TST-005 - Critical invariant: Learning cannot silently modify constitutional meaning, authority, protected behavior, or authoritative live-effect state. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-10-TST-006 - Failure containment: Inject training-data poisoning, pattern overgeneralization, silent promotion, feedback loop, unbounded self-modification. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-10-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-10-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-10-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-10-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 11, 12, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Learning, Pattern, Wisdom, Experience Ledger. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-11 - Temporal Service

Final v1.0 | Book II Chapters 13 | Component certification: PASS

Purpose and scope

Provide bounded time, ordering, freshness, lease, deadline, temporal commitment, and causality services without treating wall-clock time as authoritative distributed ordering state.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 13. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Logical Time, Wall-Clock Time, Causation Identifier.

Authoritative state

Time Facts; Logical clocks; lease records; time-source assessments; Temporal Commitment Records; scheduling constraints; deadline and freshness policies. CMP-11 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

work placement; event ownership; business scheduling authority; authority decisions. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Ordering and freshness claims always state scope, mechanism, uncertainty, and source health.

Lifecycle

Healthy > Degraded > Unsynchronized > Recovering > Healthy. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain clock regression, split-brain lease, expired authority use, causality ambiguity, time-source compromise. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Time Facts.
- Own and protect mutation of Logical clocks.
- Own and protect mutation of lease records.
- Own and protect mutation of time-source assessments.
- Own and protect mutation of Temporal Commitment Records.
- Own and protect mutation of scheduling constraints.
- Own and protect mutation of deadline and freshness policies.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-11-IF-01 / Command	Issue Lease	Command 'Issue Lease' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-02 / Command	Renew Lease	Command 'Renew Lease' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-03 / Command	Record Logical Time	Command 'Record Logical Time' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-04 / Command	Declare Time Source Degraded	Command 'Declare Time Source Degraded' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-11-IF-05 / Query	Get Current Time Context	Query `Get Current Time Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-06 / Query	Compare Causality	Query `Compare Causality` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-07 / Query	Get Freshness	Query `Get Freshness` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-08 / Event	Lease Issued	Event `Lease Issued` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-09 / Event	Lease Expired	Event `Lease Expired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-11-IF-10 / Event	Time Source Degraded	Event `Time Source Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-11-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-11-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-11-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-11-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-11-TST-005 - Critical invariant: Ordering and freshness claims always state scope, mechanism, uncertainty, and source health. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-11-TST-006 - Failure containment: Inject clock regression, split-brain lease, expired authority use, causality ambiguity, time-source compromise. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-11-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-11-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-11-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-11-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 13, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Logical Time, Wall-Clock Time, Causation Identifier. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-12 - Presence and Embodiment Layer

Final v1.0 | Book II Chapters 14, 31 | Component certification: PASS

Purpose and scope

Own bounded Presence, embodiment, audience context, and handoff state while preserving HAL's single identity.

Authority and sources

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 14, 31. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Presence, Embodiment.

Authoritative state

Presence Objects; Embodiment Bindings; Audience Context; Presence Handoffs; modality availability. CMP-12 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

HAL identity; Interaction Sessions; credentials; device ownership. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Every Presence is a bounded manifestation of one HAL identity and cannot independently hold Owner authority.

Lifecycle

Proposed > Active > Handoff > Suspended > Ended. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain presence duplication, audience confusion, sensor spoofing, handoff loss, context leakage. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Presence Objects.
- Own and protect mutation of Embodiment Bindings.
- Own and protect mutation of Audience Context.
- Own and protect mutation of Presence Handoffs.
- Own and protect mutation of modality availability.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-12-IF-01 / Command	Create Presence	Command `Create Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-02 / Command	Bind Embodiment	Command `Bind Embodiment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-03 / Command	Handoff Presence	Command `Handoff Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-04 / Command	End Presence	Command `End Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-05 / Query	Get Presence Context	Query `Get Presence Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-06 / Query	List Embodiments	Query `List Embodiments` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-07 / Query	Get Audience Context	Query `Get Audience Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-08 / Event	Presence Created	Event `Presence Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-09 / Event	Embodiment Bound	Event `Embodiment Bound` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-12-IF-10 / Event	Presence Handed Off	Event `Presence Handed Off` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-12-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-12-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-12-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-12-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-12-TST-005 - Critical invariant: Every Presence is a bounded manifestation of one HAL identity and cannot independently hold Owner authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-12-TST-006 - Failure containment: Inject presence duplication, audience confusion, sensor spoofing, handoff loss, context leakage. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-12-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-12-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-12-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-12-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 35, 40, 47-50, and 58, Book II Chapters 14, 31, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Presence, Embodiment. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-13 - Capability Registry

Final v1.0 | Book II Chapters 15 | Component certification: PASS

Purpose and scope

Own implementation-independent Capability Contracts, provider registrations, adapters, health qualifications, and selection constraints.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 15. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Capability, Capability Contract, Provider, Adapter.

Authoritative state

Capability Contracts; Provider Records; Adapter Records; qualification and health metadata. CMP-13 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

permission; authority; work placement; provider execution state. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Capability availability, provider health, or credential possession never implies Permission or Authority.

Lifecycle

Proposed > Qualified > Active > Degraded > Retired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain semantic mismatch, provider substitution, stale qualification, adapter bypass, capability shadowing. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Capability Contracts.
- Own and protect mutation of Provider Records.
- Own and protect mutation of Adapter Records.
- Own and protect mutation of qualification and health metadata.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-13-IF-01 / Command	Register Capability	Command 'Register Capability' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-02 / Command	Register Provider	Command 'Register Provider' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-03 / Command	Qualify Provider	Command 'Qualify Provider' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-04 / Command	Retire Adapter	Command 'Retire Adapter' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-05 / Query	Discover Capability	Query 'Discover Capability' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-06 / Query	Get Provider Qualification	Query 'Get Provider Qualification' MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-07 / Query	Resolve Adapter	Query `Resolve Adapter` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-08 / Event	Capability Registered	Event `Capability Registered` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-09 / Event	Provider Qualified	Event `Provider Qualified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-13-IF-10 / Event	Provider Quarantined	Event `Provider Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-13-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-13-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-13-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-13-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-13-TST-005 - Critical invariant: Capability availability, provider health, or credential possession never implies Permission or Authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-13-TST-006 - Failure containment: Inject semantic mismatch, provider substitution, stale qualification, adapter bypass, capability shadowing. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-13-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-13-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-13-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-13-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 15, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Capability, Capability Contract, Provider, Adapter. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-14 - Action and Transaction Engine

Final v1.0 | Book II Chapters 16 | Component certification: PASS

Purpose and scope

Coordinate authorized Actions and Transactions across prepare, commit, observation, rollback, and compensation boundaries.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 16. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Action, Transaction, Commit Barrier, Rollback, Compensation, Reality Boundary.

Authoritative state

Action Records; Transaction Records; Commit Barrier decisions; idempotency records; rollback and compensation state. CMP-14 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

intent; authority grants; external-domain policy; outcome evaluation. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

No real effect crosses the Commit Barrier without current authority, verification, exact state, evidence, and Reality Boundary approval.

Lifecycle

Proposed > Authorized > Prepared > Committed > Completed; failure to Rolled Back or Compensating. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain duplicate effect, partial commit, stale Permission decision, rollback lie, uncontained external effect. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Action Records.
- Own and protect mutation of Transaction Records.
- Own and protect mutation of Commit Barrier decisions.
- Own and protect mutation of idempotency records.
- Own and protect mutation of rollback and compensation state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-14-IF-01 / Command	Propose Action	Command `Propose Action` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-02 / Command	Prepare Transaction	Command `Prepare Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-03 / Command	Commit Transaction	Command `Commit Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-04 / Command	Rollback Transaction	Command `Rollback Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-05 / Command	Begin Compensation	Command `Begin Compensation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-06 / Query	Get Transaction State	Query `Get Transaction State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-07 / Query	Explain Commit Decision	Query `Explain Commit Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-08 / Query	Get Idempotency Result	Query `Get Idempotency Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-09 / Event	Action Authorized	Event `Action Authorized` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-10 / Event	Transaction Committed	Event `Transaction Committed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-11 / Event	Transaction Rolled Back	Event `Transaction Rolled Back` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-14-IF-12 / Event	Compensation Started	Event `Compensation Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-14-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-14-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-14-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-14-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-14-TST-005 - Critical invariant: No real effect crosses the Commit Barrier without current authority, verification, exact state, evidence, and Reality Boundary approval. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-14-TST-006 - Failure containment: Inject duplicate effect, partial commit, stale Permission decision, rollback lie, uncontained external effect. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-14-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-14-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-14-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-14-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 16, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Action, Transaction, Commit Barrier, Rollback, Compensation, Reality Boundary. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-15 - Verification Service

Final v1.0 | Book II Chapters 17, 35 | Component certification: PASS

Purpose and scope

Evaluate explicit claims against declared criteria, methods, environments, evidence, uncertainty, and reproducibility requirements.

Authority and sources

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 17, 35. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Verification Plan, Assurance Case, Certification, Confidence.

Authoritative state

Verification Plans; Verification Results; claim-evaluation state; fidelity and confidence records. CMP-15 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Evidence Objects; Certification decisions; simulation runtime; release approval. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Verification is scoped and evidence-based; it never converts an unsupported claim into authority or certification.

Lifecycle

Proposed > Running > Verified, Falsified, Inconclusive, or Invalidated. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain criterion drift, environment mismatch, evidence substitution, irreproducible result, confidence inflation. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Verification Plans.
- Own and protect mutation of Verification Results.
- Own and protect mutation of claim-evaluation state.
- Own and protect mutation of fidelity and confidence records.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-15-IF-01 / Command	Start Verification	Command `Start Verification` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-02 / Command	Record Verification Result	Command `Record Verification Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-03 / Command	Invalidate Result	Command `Invalidate Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-04 / Command	Request Reproduction	Command `Request Reproduction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-05 / Query	Get Verification Result	Query `Get Verification Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-06 / Query	Explain Confidence	Query `Explain Confidence` MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-07 / Query	List Defeaters	Query `List Defeaters` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-08 / Event	Verification Started	Event `Verification Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-09 / Event	Claim Verified	Event `Claim Verified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-10 / Event	Claim Falsified	Event `Claim Falsified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-15-IF-11 / Event	Result Invalidated	Event `Result Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-15-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-15-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-15-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-15-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-15-TST-005 - Critical invariant: Verification is scoped and evidence-based; it never converts an unsupported claim into authority or certification. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-15-TST-006 - Failure containment: Inject criterion drift, environment mismatch, evidence substitution, irreproducible result, confidence inflation. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-15-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-15-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-15-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-15-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.

- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 24-29, 35, 40-43, 47-50, and 58, Book II Chapters 17, 35, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Verification Plan, Assurance Case, Certification, Confidence. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-16 - Simulation and Digital Twin Platform

Final v1.0 | Book II Chapters 17 | Component certification: PASS

Purpose and scope

Run isolated simulation, Digital Twin, counterfactual, failure-injection, and shadow workloads without ungoverned real effects.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 17. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Simulation, Digital Twin, Shadow Execution, Canary.

Authoritative state

Simulation Runs; Digital Twin State; fidelity models; scenario and isolation records. CMP-16 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

authoritative live-effect state; Certification; real-world commit; Evidence admission. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Simulation, Digital Twin, and Shadow Execution remain technically incapable of ungoverned real effects.

Lifecycle

Draft > Ready > Running > Completed, Failed, or Quarantined. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain reality leakage, stale twin, fidelity misstatement, unsafe fixture, resource escape. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Simulation Runs.
- Own and protect mutation of Digital Twin State.
- Own and protect mutation of fidelity models.
- Own and protect mutation of scenario and isolation records.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-16-IF-01 / Command	Create Simulation	Command `Create Simulation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-02 / Command	Load Twin State	Command `Load Twin State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-03 / Command	Run Scenario	Command `Run Scenario` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-04 / Command	Inject Failure	Command `Inject Failure` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-05 / Command	Terminate Simulation	Command `Terminate Simulation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-06 / Query	Get Scenario Result	Query `Get Scenario Result` MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-07 / Query	Get Fidelity Score	Query `Get Fidelity Score` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-08 / Query	Get Isolation Status	Query `Get Isolation Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-09 / Event	Simulation Started	Event `Simulation Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-10 / Event	Scenario Completed	Event `Scenario Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-16-IF-11 / Event	Isolation Breach Detected	Event `Isolation Breach Detected` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-16-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-16-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-16-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-16-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-16-TST-005 - Critical invariant: Simulation, Digital Twin, and Shadow Execution remain technically incapable of ungoverned real effects. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-16-TST-006 - Failure containment: Inject reality leakage, stale twin, fidelity misstatement, unsafe fixture, resource escape. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-16-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-16-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-16-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-16-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.

- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 17, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Simulation, Digital Twin, Shadow Execution, Canary. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-17 - Trust Service

Final v1.0 | Book II Chapters 18 | Component certification: PASS

Purpose and scope

Produce multidimensional, evidence-based, scoped, expiring Trust Assessments without granting authority or permission.

Authority and sources

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 18. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Trust, Trust Domain, Trust Boundary.

Authoritative state

Trust Assessments; trust dimensions; confidence and expiry metadata; trust-policy inputs. CMP-17 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Evidence Objects; Authority; Permission; Treaties; authoritative Identity state. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Trust may inform policy but never grants Authority, Permission, identity, or Treaty scope.

Lifecycle

Requested > Assessed > Degraded > Expired or Revoked. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain trust-as-authority, stale evidence, dimension collapse, reputation contagion, expiry bypass. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Trust Assessments.
- Own and protect mutation of trust dimensions.
- Own and protect mutation of confidence and expiry metadata.
- Own and protect mutation of trust-policy inputs.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-17-IF-01 / Command	Request Trust Assessment	Command `Request Trust Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-02 / Command	Update Trust Dimension	Command `Update Trust Dimension` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-03 / Command	Invalidate Assessment	Command `Invalidate Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-04 / Query	Get Trust Assessment	Query `Get Trust Assessment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-05 / Query	Explain Trust Basis	Query `Explain Trust Basis` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-06 / Query	List Expiring Assessments	Query `List Expiring Assessments` MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-07 / Event	Trust Assessed	Event `Trust Assessed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-08 / Event	Trust Degraded	Event `Trust Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-17-IF-09 / Event	Assessment Expired	Event `Assessment Expired` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-17-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-17-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-17-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-17-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-17-TST-005 - Critical invariant: Trust may inform policy but never grants Authority, Permission, identity, or Treaty scope. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-17-TST-006 - Failure containment: Inject trust-as-authority, stale evidence, dimension collapse, reputation contagion, expiry bypass. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-17-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-17-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-17-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-17-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 24-29, 35, 40-43, 47-50, and 58, Book II Chapters 18, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Trust, Trust Domain, Trust Boundary. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-18 - Evidence Service

Final v1.0 | Book II Chapters 18, 25 | Component certification: PASS

Purpose and scope

Admit and govern immutable Evidence Objects with provenance, custody, signatures, verification state, confidence, domain, and expiry.

Authority and sources

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 18, 25. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Evidence Candidate, Evidence Service, Chain of Custody, Audit Record.

Authoritative state

Evidence Objects; custody chains; admission decisions; signature and verification state. CMP-18 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Audit Records; telemetry; claims; Knowledge Graph; Trust Assessments. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

An admitted Evidence Object is immutable; correction occurs only through linked later objects.

Lifecycle

Candidate > Admitted > Challenged > Superseded or Expired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain custody break, provenance forgery, mutable evidence, signature failure, unauthorized disclosure. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Evidence Objects.
- Own and protect mutation of custody chains.
- Own and protect mutation of admission decisions.
- Own and protect mutation of signature and verification state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-18-IF-01 / Command	Submit Evidence Candidate	Command `Submit Evidence Candidate` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-02 / Command	Admit Evidence Object	Command `Admit Evidence Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-03 / Command	Challenge Evidence	Command `Challenge Evidence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-04 / Command	Record Supersession	Command `Record Supersession` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-05 / Query	Get Evidence Object	Query `Get Evidence Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-06 / Query	Trace Custody	Query `Trace Custody` MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-07 / Query	Verify Evidence Integrity	Query `Verify Evidence Integrity` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-08 / Event	Evidence Admitted	Event `Evidence Admitted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-09 / Event	Evidence Challenged	Event `Evidence Challenged` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-18-IF-10 / Event	Evidence Superseded	Event `Evidence Superseded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-18-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-18-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-18-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-18-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-18-TST-005 - Critical invariant: An admitted Evidence Object is immutable; correction occurs only through linked later objects. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-18-TST-006 - Failure containment: Inject custody break, provenance forgery, mutable evidence, signature failure, unauthorized disclosure. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-18-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-18-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-18-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-18-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.

- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 24-29, 35, 40-43, 47-50, and 58, Book II Chapters 18, 25, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Evidence Candidate, Evidence Service, Chain of Custody, Audit Record. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-19 - Privacy and Data Governance Service

Final v1.0 | Book II Chapters 19 | Component certification: PASS

Purpose and scope

Own data classification, purpose, authority, minimization, consent, retention, disclosure, deletion, and inference-risk policy state.

Authority and sources

Book I: Decisions 24-29, 35, 40-43, 47-50, and 58. Book II: Chapters 19. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Personal Data, Sensitive Data, Purpose Limitation, Data Minimization, Retention Class.

Authoritative state

Data Classification Records; purpose bindings; retention classes; consent and authority records; deletion obligations. CMP-19 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

source data contents; authoritative Identity state; Treaty lifecycle; application business logic. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Data use is permitted only for declared authorized purpose with minimization, retention, disclosure, and disposal controls.

Lifecycle

Unclassified > Classified > Active > Restricted > Disposed. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain purpose drift, overcollection, retention breach, inference leakage, deletion incompleteness. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Data Classification Records.
- Own and protect mutation of purpose bindings.
- Own and protect mutation of retention classes.
- Own and protect mutation of consent and authority records.
- Own and protect mutation of deletion obligations.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-19-IF-01 / Command	Classify Data	Command `Classify Data` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-02 / Command	Authorize Purpose	Command `Authorize Purpose` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-03 / Command	Set Retention	Command `Set Retention` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-04 / Command	Order Deletion	Command `Order Deletion` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-05 / Command	Restrict Disclosure	Command `Restrict Disclosure` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-06 / Query	Get Data Policy	Query `Get Data Policy` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-07 / Query	Evaluate Data Use	Query `Evaluate Data Use` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-08 / Query	Get Deletion Status	Query `Get Deletion Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-09 / Event	Data Classified	Event `Data Classified` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-10 / Event	Purpose Authorized	Event `Purpose Authorized` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-11 / Event	Deletion Completed	Event `Deletion Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-19-IF-12 / Event	Use Denied	Event `Use Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-19-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-19-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-19-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-19-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-19-TST-005 - Critical invariant: Data use is permitted only for declared authorized purpose with minimization, retention, disclosure, and disposal controls. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-19-TST-006 - Failure containment: Inject purpose drift, overcollection, retention breach, inference leakage, deletion incompleteness. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-19-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-19-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-19-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-19-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 24-29, 35, 40-43, 47-50, and 58, Book II Chapters 19, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Personal Data, Sensitive Data, Purpose Limitation, Data Minimization, Retention Class. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-20 - Constitutional Firewall

Final v1.0 | Book II Chapters 20, 21 | Component certification: PASS

Purpose and scope

Enforce cross-domain identity, data, capability, authority, and evidence exchange against active signed Treaty decisions.

Authority and sources

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 20, 21. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony, Constitutional Firewall, External Trust Domain, Treaty.

Authoritative state

firewall decisions; cross-domain session state; Treaty enforcement cache; exchange audit bindings; cross-domain incident records. CMP-20 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Treaty proposals or approval; native authority grants; external authoritative Identity state; payload source state. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

No external exchange bypasses the Constitutional Firewall, and a Treaty cannot authorize what Book I prohibits.

Lifecycle

Closed > Negotiating > Active > Restricted > Closed. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain Treaty mismatch, identity substitution, policy-cache staleness, exfiltration attempt, domain compromise. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of firewall decisions.
- Own and protect mutation of cross-domain session state.
- Own and protect mutation of Treaty enforcement cache.
- Own and protect mutation of exchange audit bindings.
- Own and protect mutation of cross-domain incident records.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-20-IF-01 / Command	Open Domain Session	Command `Open Domain Session` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-02 / Command	Authorize Exchange	Command `Authorize Exchange` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-03 / Command	Terminate Exchange	Command `Terminate Exchange` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-04 / Command	Quarantine Payload	Command `Quarantine Payload` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-05 / Query	Get Firewall Decision	Query `Get Firewall Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-06 / Query	Get Active Treaty View	Query `Get Active Treaty View` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-07 / Query	Explain Denial	Query `Explain Denial` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-08 / Event	Exchange Allowed	Event `Exchange Allowed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-09 / Event	Exchange Denied	Event `Exchange Denied` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-20-IF-10 / Event	Domain Session Terminated	Event `Domain Session Terminated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-20-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-20-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-20-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-20-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-20-TST-005 - Critical invariant: No external exchange bypasses the Constitutional Firewall, and a Treaty cannot authorize what Book I prohibits. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-20-TST-006 - Failure containment: Inject Treaty mismatch, identity substitution, policy-cache staleness, exfiltration attempt, domain compromise. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-20-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-20-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-20-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-20-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.

- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58, Book II Chapters 20, 21, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony, Constitutional Firewall, External Trust Domain, Treaty. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-21 - Treaty Manager

Final v1.0 | Book II Chapters 21 | Component certification: PASS

Purpose and scope

Own exact, scoped, time-bounded, revocable, auditable, explicitly Owner-authorized Treaty proposals, approvals, lifecycle, and termination.

Authority and sources

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 21. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Treaty, External Trust Domain, Owner Authorization Ceremony.

Authoritative state

Treaty Records; proposal versions; Owner Authorization Ceremony bindings; activation, suspension, and revocation state. CMP-21 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

cross-domain enforcement; Trust Assessments; Owner identity; ordinary delegation. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Activation requires an Owner Authorization Ceremony bound to the exact Treaty; conversation or ordinary delegation is insufficient.

Lifecycle

Draft > Reviewed > Owner Authorized > Active > Suspended or Revoked or Expired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain Owner Authorization Ceremony mismatch, scope ambiguity, expired Treaty use, revocation lag, external-domain repudiation. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Treaty Records.
- Own and protect mutation of proposal versions.
- Own and protect mutation of Owner Authorization Ceremony bindings.
- Own and protect mutation of activation, suspension, and revocation state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-21-IF-01 / Command	Propose Treaty	Command 'Propose Treaty' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-02 / Command	Authorize Treaty	Command 'Authorize Treaty' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-03 / Command	Activate Treaty	Command 'Activate Treaty' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-04 / Command	Suspend Treaty	Command 'Suspend Treaty' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-05 / Command	Revoke Treaty	Command 'Revoke Treaty' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-06 / Query	Get Treaty	Query 'Get Treaty' MUST carry identity, authority context where applicable,

		schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-07 / Query	Compare Treaty Version	Query `Compare Treaty Version` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-08 / Query	List Active Treaties	Query `List Active Treaties` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-09 / Event	Treaty Proposed	Event `Treaty Proposed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-10 / Event	Treaty Activated	Event `Treaty Activated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-11 / Event	Treaty Suspended	Event `Treaty Suspended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-21-IF-12 / Event	Treaty Revoked	Event `Treaty Revoked` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-21-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-21-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-21-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-21-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-21-TST-005 - Critical invariant: Activation requires an Owner Authorization Ceremony bound to the exact Treaty; conversation or ordinary delegation is insufficient. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-21-TST-006 - Failure containment: Inject Owner Authorization Ceremony mismatch, scope ambiguity, expired Treaty use, revocation lag, external-domain repudiation. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-21-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-21-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-21-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-21-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58, Book II Chapters 21, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Treaty, External Trust Domain, Owner Authorization Ceremony. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-22 - Event and Messaging Platform

Final v1.0 | Book II Chapters 22, 23 | Component certification: PASS

Purpose and scope

Provide durable commands, queries, events, envelopes, ordering scopes, deduplication, outbox, thread-delivery, and delivery semantics.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 22, 23. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Command, Query, Event, Message Envelope, Event Journal, Transactional Outbox.

Authoritative state

Event Journal; message envelopes; deduplication records; delivery and subscription state; Thread Delivery Records; Transactional Outbox publication state. CMP-22 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

domain authoritative state; command Permission evaluation; global ordering; business process ownership. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Commands request possible change, Events record completed facts, and Queries do not intentionally mutate authoritative state.

Lifecycle

Created > Accepted > Delivered > Acknowledged or Failed or Expired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain duplicate delivery, causation loss, poison message, ordering overclaim, outbox divergence. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Event Journal.
- Own and protect mutation of message envelopes.
- Own and protect mutation of deduplication records.
- Own and protect mutation of delivery and subscription state.
- Own and protect mutation of Thread Delivery Records.
- Own and protect mutation of Transactional Outbox publication state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-22-IF-01 / Command	Publish Command	Command `Publish Command` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-02 / Command	Append Event	Command `Append Event` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-03 / Command	Acknowledge Delivery	Command `Acknowledge Delivery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-04 / Command	Quarantine Message	Command `Quarantine Message` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-05 / Query	Query Event Stream	Query `Query Event Stream` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-06 / Query	Get Delivery State	Query `Get Delivery State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-07 / Query	Trace Causation	Query `Trace Causation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-08 / Event	Message Accepted	Event `Message Accepted` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-09 / Event	Event Published	Event `Event Published` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-10 / Event	Delivery Failed	Event `Delivery Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-22-IF-11 / Event	Message Quarantined	Event `Message Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-22-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-22-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-22-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-22-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-22-TST-005 - Critical invariant: Commands request possible change, Events record completed facts, and Queries do not intentionally mutate authoritative state. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-22-TST-006 - Failure containment: Inject duplicate delivery, causation loss, poison message, ordering overclaim, outbox divergence. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-22-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-22-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-22-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-22-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 22, 23, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Command, Query, Event, Message Envelope, Event Journal, Transactional Outbox. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-23 - State and Persistence Platform

Final v1.0 | Book II Chapters 24 | Component certification: PASS

Purpose and scope

Provide durable authoritative storage primitives, concurrency controls, snapshots, projections, replicas, and migration facilities while preserving domain ownership.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 24. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, State, Authoritative State, Projection, Replica.

Authoritative state

storage engines; commit logs; snapshots; replication metadata; projection and migration state. CMP-23 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

semantic ownership of domain records; policy decisions; evidence meaning; global transaction authority. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Each authoritative state domain has exactly one mutation owner; replicas, caches, and projections remain explicitly derived.

Lifecycle

Provisioning > Ready > Degraded > Recovering > Ready or Retired. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain split brain, lost update, snapshot corruption, migration partial failure, projection treated as authority. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of storage engines.
- Own and protect mutation of commit logs.
- Own and protect mutation of snapshots.
- Own and protect mutation of replication metadata.
- Own and protect mutation of projection and migration state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-23-IF-01 / Command	Commit State Change	Command `Commit State Change` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-02 / Command	Create Snapshot	Command `Create Snapshot` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-03 / Command	Migrate Store	Command `Migrate Store` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-04 / Command	Rebuild Projection	Command `Rebuild Projection` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-05 / Query	Read State	Query `Read State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit

		success or denial semantics. Book IX owns wire details.
CMP-23-IF-06 / Query	Get Version	Query `Get Version` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-07 / Query	Verify Replica	Query `Verify Replica` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-08 / Query	Get Migration State	Query `Get Migration State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-09 / Event	State Committed	Event `State Committed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-10 / Event	Snapshot Created	Event `Snapshot Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-11 / Event	Migration Completed	Event `Migration Completed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-23-IF-12 / Event	Replica Diverged	Event `Replica Diverged` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-23-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-23-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-23-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-23-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-23-TST-005 - Critical invariant: Each authoritative state domain has exactly one mutation owner; replicas, caches, and projections remain explicitly derived. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-23-TST-006 - Failure containment: Inject split brain, lost update, snapshot corruption, migration partial failure, projection treated as authority. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-23-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-23-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-23-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-23-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 24, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, State, Authoritative State, Projection, Replica. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-24 - Observability and Audit Platform

Final v1.0 | Book II Chapters 25 | Component certification: PASS

Purpose and scope

Collect structured logs, metrics, traces, Audit Records, health evidence candidates, and tamper-evident audit chronology without owning Evidence Objects.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 25. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Audit Record, Evidence Candidate.

Authoritative state

Audit Ledger; Audit Records; telemetry pipelines; metric and trace stores; alert state. CMP-24 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Evidence Objects; domain authoritative state; Trust Assessments; incident command. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Observability may produce Evidence Candidates, but only Evidence Service admits Evidence Objects or changes evidentiary meaning.

Lifecycle

Healthy > Degraded > Buffering > Recovering > Healthy. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain sensitive logging, audit gap, trace spoofing, telemetry loss, ledger tampering. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Audit Ledger.
- Own and protect mutation of Audit Records.
- Own and protect mutation of telemetry pipelines.
- Own and protect mutation of metric and trace stores.
- Own and protect mutation of alert state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-24-IF-01 / Command	Append Audit Record	Command `Append Audit Record` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-02 / Command	Emit Telemetry	Command `Emit Telemetry` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-03 / Command	Declare Alert	Command `Declare Alert` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-04 / Command	Seal Audit Segment	Command `Seal Audit Segment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-05 / Query	Query Audit Ledger	Query `Query Audit Ledger` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-06 / Query	Get Trace	Query `Get Trace` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-07 / Query	Get Metric	Query `Get Metric` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-08 / Query	Verify Audit Chain	Query `Verify Audit Chain` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-09 / Event	Audit Record Appended	Event `Audit Record Appended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-10 / Event	Alert Raised	Event `Alert Raised` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-24-IF-11 / Event	Audit Chain Broken	Event `Audit Chain Broken` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-24-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-24-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-24-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-24-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-24-TST-005 - Critical invariant: Observability may produce Evidence Candidates, but only Evidence Service admits Evidence Objects or changes evidentiary meaning. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-24-TST-006 - Failure containment: Inject sensitive logging, audit gap, trace spoofing, telemetry loss, ledger tampering. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-24-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-24-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-24-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-24-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 25, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Audit Record, Evidence Candidate. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-25 - Runtime Supervisor

Final v1.0 | Book II Chapters 02, 27, 29, 34 | Component certification: PASS

Purpose and scope

Control runtime mode, service lifecycle, readiness, health, resource limits, deployment instances, degraded modes, quarantine, and restart policy.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 02, 27, 29, 34. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Runtime, Service, Supervisor, Readiness, Health, Liveness.

Authoritative state

Current Runtime Mode; Service Instance Records; Desired and Observed State; health and readiness state; resource assignments; quarantine state. CMP-25 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

application domain state; work admission; recovery authority; release certification. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Restart, locality, or process liveness never establishes readiness, correctness, authority, or recovery.

Lifecycle

Provisioned > Starting > Ready > Degraded > Quarantined > Recovering > Ready or Stopped. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain restart loop, health false positive, resource exhaustion, quarantine bypass, configuration drift. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Current Runtime Mode.
- Own and protect mutation of Service Instance Records.
- Own and protect mutation of Desired and Observed State.
- Own and protect mutation of health and readiness state.
- Own and protect mutation of resource assignments.
- Own and protect mutation of quarantine state.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-25-IF-01 / Command	Start Service	Command `Start Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-02 / Command	Stop Service	Command `Stop Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-03 / Command	Restart Service	Command `Restart Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-04 / Command	Enter Degraded Mode	Command `Enter Degraded Mode` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-05 / Command	Quarantine Service	Command `Quarantine Service` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-06 / Query	Get Service State	Query `Get Service State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-07 / Query	Get Health	Query `Get Health` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-08 / Query	Get Resource Use	Query `Get Resource Use` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-09 / Event	Service Ready	Event `Service Ready` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-10 / Event	Service Degraded	Event `Service Degraded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-11 / Event	Service Quarantined	Event `Service Quarantined` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-25-IF-12 / Event	Service Stopped	Event `Service Stopped` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-25-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-25-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-25-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-25-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-25-TST-005 - Critical invariant: Restart, locality, or process liveness never establishes readiness, correctness, authority, or recovery. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-25-TST-006 - Failure containment: Inject restart loop, health false positive, resource exhaustion, quarantine bypass, configuration drift. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-25-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-25-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-25-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-25-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 02, 27, 29, 34, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Runtime, Service, Supervisor, Readiness, Health, Liveness. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-26 - Recovery Coordinator

Final v1.0 | Book II Chapters 27, 28 | Component certification: PASS

Purpose and scope

Coordinate containment, recovery plans, identity and state reconciliation, restoration, rollback, compensation, and continuity evidence.

Authority and sources

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 27, 28. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Recovery, Recovery Point Objective, Recovery Time Objective, Quarantine.

Authoritative state

Recovery Cases; recovery plans; reconciliation decisions; restoration progress; RPO and RTO evidence. CMP-26 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

domain authoritative state; constitutional admission; incident command; backup storage. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Recovery restores verified identity, authority, state, evidence, and trust; restart alone is not Recovery.

Lifecycle

Detected > Contained > Planned > Restoring > Validating > Closed or Escalated. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain wrong recovery point, identity discontinuity, partition conflict, evidence loss, premature service return. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Recovery Cases.
- Own and protect mutation of recovery plans.
- Own and protect mutation of reconciliation decisions.
- Own and protect mutation of restoration progress.
- Own and protect mutation of RPO and RTO evidence.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-26-IF-01 / Command	Open Recovery Case	Command `Open Recovery Case` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-02 / Command	Authorize Recovery Step	Command `Authorize Recovery Step` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-03 / Command	Restore State	Command `Restore State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-04 / Command	Reconcile Partition	Command `Reconcile Partition` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-05 / Command	Close Recovery	Command `Close Recovery` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance,

		and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-06 / Query	Get Recovery State	Query `Get Recovery State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-07 / Query	Get RPO/RTO Result	Query `Get RPO/RTO Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-08 / Query	Explain Reconciliation	Query `Explain Reconciliation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-09 / Event	Recovery Started	Event `Recovery Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-10 / Event	State Restored	Event `State Restored` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-11 / Event	Partition Reconciled	Event `Partition Reconciled` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-26-IF-12 / Event	Recovery Failed	Event `Recovery Failed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-26-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-26-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-26-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-26-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-26-TST-005 - Critical invariant: Recovery restores verified identity, authority, state, evidence, and trust; restart alone is not Recovery. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-26-TST-006 - Failure containment: Inject wrong recovery point, identity discontinuity, partition conflict, evidence loss, premature service return. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-26-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-26-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-26-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-26-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 35, 38-43, 47-51, and 58, Book II Chapters 27, 28, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Recovery, Recovery Point Objective, Recovery Time Objective, Quarantine. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-27 - Human Interaction Layer

Final v1.0 | Book II Chapters 14, 31 | Component certification: PASS

Purpose and scope

Own Interaction Sessions, modality coordination, accessibility, consent cues, clarification, confirmation, and human-visible action status.

Authority and sources

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 14, 31. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Presence, Principal.

Authoritative state

Interaction Sessions; turn and modality state; clarification and confirmation records; accessibility preferences. CMP-27 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Presence Objects; human Identity Records; Intent; Authority; action execution. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Human-facing language, convenience, or conversational assent never substitutes for governed identity, Authority, or the exact protected Permission decision.

Lifecycle

Proposed > Active > Awaiting Clarification or Confirmation > Completed or Abandoned. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain identity confusion, dark pattern, inaccessible output, consent ambiguity, stale action status. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Interaction Sessions.
- Own and protect mutation of turn and modality state.
- Own and protect mutation of clarification and confirmation records.
- Own and protect mutation of accessibility preferences.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-27-IF-01 / Command	Start Interaction	Command `Start Interaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-02 / Command	Submit Human Input	Command `Submit Human Input` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-03 / Command	Request Confirmation	Command `Request Confirmation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-04 / Command	End Interaction	Command `End Interaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-05 / Query	Get Session State	Query `Get Session State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-06 / Query	Get Accessible Representation	Query `Get Accessible Representation` MUST carry identity, authority context

		where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-07 / Query	Get Action Status	Query `Get Action Status` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-08 / Event	Interaction Started	Event `Interaction Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-09 / Event	Clarification Requested	Event `Clarification Requested` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-10 / Event	Confirmation Recorded	Event `Confirmation Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-27-IF-11 / Event	Interaction Ended	Event `Interaction Ended` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-27-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-27-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-27-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-27-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-27-TST-005 - Critical invariant: Human-facing language, convenience, or conversational assent never substitutes for governed identity, Authority, or the exact protected Permission decision. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-27-TST-006 - Failure containment: Inject identity confusion, dark pattern, inaccessible output, consent ambiguity, stale action status. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-27-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-27-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-27-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-27-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 35, 40, 47-50, and 58, Book II Chapters 14, 31, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Presence, Principal. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-28 - Self-Model and Constitutional Mirror

Final v1.0 | Book II Chapters 30, 33 | Component certification: PASS

Purpose and scope

Maintain evidence-linked descriptions of HAL identity, capabilities, limitations, governing constraints, and conformance without becoming self-authorizing.

Authority and sources

Book I: Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58. Book II: Chapters 30, 33. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony, Self Model, Constitutional Mirror, Semantic Change.

Authoritative state

Self-Model Records; Constitutional Mirror versions; capability and limitation projections; conformance summaries. CMP-28 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Constitution; constitutional authority; Capability Contracts; certification decisions; source evidence. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Self-description is descriptive, evidence-linked, and non-self-authorizing; it cannot create identity, authority, or capability.

Lifecycle

Draft > Verified > Published > Superseded or Withdrawn. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain self-issued Authority or Permission, stale capability claim, missing limitation, evidence disconnect, identity fragmentation. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Self-Model Records.
- Own and protect mutation of Constitutional Mirror versions.
- Own and protect mutation of capability and limitation projections.
- Own and protect mutation of conformance summaries.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-28-IF-01 / Command	Refresh Self Model	Command 'Refresh Self Model' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-02 / Command	Publish Constitutional Mirror	Command 'Publish Constitutional Mirror' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-03 / Command	Invalidate Projection	Command 'Invalidate Projection' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-04 / Query	Get Self Description	Query 'Get Self Description' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-05 / Query	Trace Mirror Claim	Query 'Trace Mirror Claim' MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-06 / Query	Get Declared Limitation	Query 'Get Declared Limitation' MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-07 / Event	Self Model Refreshed	Event `Self Model Refreshed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-08 / Event	Mirror Published	Event `Mirror Published` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-28-IF-09 / Event	Projection Invalidated	Event `Projection Invalidated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-28-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-28-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-28-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-28-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-28-TST-005 - Critical invariant: Self-description is descriptive, evidence-linked, and non-self-authorizing; it cannot create identity, authority, or capability. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-28-TST-006 - Failure containment: Inject self-issued Authority or Permission, stale capability claim, missing limitation, evidence disconnect, identity fragmentation. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-28-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-28-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-28-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-28-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
- No trust, role, identity, credential, or capability as Authority.
- No fail-open bypass when policy, verification, evidence, or audit is unavailable.
- No Event before commit; no cache or projection as authority; no false rollback of irreversible effects.
- No Book IX machine contract embedded as independent architecture.

Traceability and completion

Twelve numbered requirements map to Book I Decisions 1-7, 24-29, 38-43, 47-51, 56, and 58, Book II Chapters 30, 33, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Constitution, Constitutional Invariant, Constitutional Kernel, Owner Authorization Ceremony, Self Model, Constitutional Mirror, Semantic Change. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

CMP-29 - Outcome and Success Evaluation Service

Final v1.0 | Book II Chapters 32 | Component certification: PASS

Purpose and scope

Evaluate Goal outcomes, side effects, constitutional costs, human impact, and declared success criteria using evidence and uncertainty.

Authority and sources

Book I: Decisions 8-23, 30-37, 40-43, 47-50, and 58. Book II: Chapters 32. Book III: Chapters 1, 3, 4, 5, 6, 7, 8, and 9; Chapter 2 for repository, dependency, configuration, and secret controls. Book X: HAL, Identity, Authority, Evidence Object, Verification, Conformance, Outcome Object, Success, Goal, Judgment.

Authoritative state

Outcome Objects; success evaluations; side-effect records; review and learning referrals. CMP-29 is the sole mutation owner; all caches, projections, replicas, and indexes remain derived.

Explicit non-responsibilities

Goals; Actions; Evidence Objects; Judgment; constitutional values. The component cannot gain authority through data, code, credentials, trust, usefulness, or locality.

Critical invariant

Success is evaluated from declared outcomes and constitutional costs, never from activity, completion, or self-reported confidence alone.

Lifecycle

Pending > Observing > Evaluated > Under Review > Closed or Reopened. Every transition requires prior version, guard, authority where applicable, idempotency, time context, and immutable evidence.

Identity, authority, and policy

Every command resolves Identity, authenticates current context, evaluates Authority and Permission, enforces delegation scope, policy, operating mode, and exact-effect binding. Trust and credentials are not Authority.

Trust, security, and privacy

All callers and dependencies remain outside the semantic trust boundary. Apply least privilege, integrity, replay resistance, secret indirection, validation, purpose limitation, minimization, retention, disclosure, deletion, and sensitive-logging prohibitions.

Failure containment and recovery

Detect and contain vanity metric, missing side effect, premature evaluation, evidence cherry-pick, goal substitution. Unsafe mutation stops; evidence is preserved. Recovery reconciles identity, state, authority, provenance, partitions, dependencies, and invariants before mutation resumes.

Observability and evidence

Emit structured admissions, denials, decisions, transitions, failures, privacy and security events, recovery, migration, and conformance results with correlation, causation, classification, redaction, integrity, retention, and time-quality metadata.

Performance and topology

Define evidence-backed latency, throughput, queue, storage, freshness, and recovery objectives before qualification. Replication and scaling cannot multiply identity, state ownership, or Authority.

Compatibility and migration

Version contracts, state, policy, events, and evidence. Test compatibility; use recoverable checkpoints, signed artifacts, staged qualification, and post-migration reconciliation.

Responsibilities

- Own and protect mutation of Outcome Objects.
- Own and protect mutation of success evaluations.
- Own and protect mutation of side-effect records.
- Own and protect mutation of review and learning referrals.

Logical interfaces and Book IX handoff

ID / kind	Operation	Semantic obligation
CMP-29-IF-01 / Command	Open Outcome Evaluation	Command `Open Outcome Evaluation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-02 / Command	Record Outcome	Command `Record Outcome` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-03 / Command	Evaluate Success	Command `Evaluate Success` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-04 / Command	Refer Learning	Command `Refer Learning` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-05 / Query	Get Outcome Object	Query `Get Outcome Object` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-06 / Query	Explain Success Result	Query `Explain Success Result` MUST carry identity, authority context where

		applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-07 / Query	List Side Effects	Query `List Side Effects` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-08 / Event	Outcome Recorded	Event `Outcome Recorded` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-09 / Event	Success Evaluated	Event `Success Evaluated` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.
CMP-29-IF-10 / Event	Review Required	Event `Review Required` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Conformance tests

CMP-29-TST-001 - Sole-owner mutation: Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-29-TST-002 - Authority denial: Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-29-TST-003 - Valid lifecycle: Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-29-TST-004 - Invalid lifecycle: Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-29-TST-005 - Critical invariant: Success is evaluated from declared outcomes and constitutional costs, never from activity, completion, or self-reported confidence alone. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-29-TST-006 - Failure containment: Inject vanity metric, missing side effect, premature evaluation, evidence cherry-pick, goal substitution. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-29-TST-007 - Recovery: Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-29-TST-008 - Contract compatibility: Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-29-TST-009 - Privacy and security: Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-29-TST-010 - Topology independence: Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Prohibited shortcuts

- No direct authoritative datastore mutation.
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Traceability and completion

Twelve numbered requirements map to Book I Decisions 8-23, 30-37, 40-43, 47-50, and 58, Book II Chapters 32, Book III common controls, and Book X terms HAL, Identity, Authority, Evidence Object, Verification, Conformance, Outcome Object, Success, Goal, Judgment. The component review closed all internally resolvable findings. No Owner Review item. Complete for Book IV v1.0.

Appendix A - Authoritative State Ownership

136 state domains are assigned to unique mutation owners. The complete matrix is maintained in the repository.

CMP-01 - Constitutional Kernel

Constitutional Identity Record; Constitution and Invariant Registry; protected-policy admission metadata; Owner Authorization Ceremony Registry; Protected Audit Anchor; Recovery Admission Record

CMP-02 - Identity and Continuity Service

Identity Records; identifier namespaces; continuity lineage; identity lifecycle; authentication-assurance bindings

CMP-03 - Authority and Delegation Service

Authority Grants; Delegation Records; revocations; Policy Decision Records; authority-context envelopes

CMP-04 - Intent Service

Intent Records; Goal and Objective hierarchy; intent version lineage; acceptance and outcome criteria

CMP-05 - Cognitive Orchestrator

Execution Graphs; orchestration runs; provider-selection records; result assembly state

CMP-06 - Attention Manager

Attention Objects; work-admission decisions; priority and preemption records; placement leases

CMP-07 - Judgment Service

Decision Objects; alternative evaluations; uncertainty assessments; review conditions

CMP-08 - Knowledge Service

Knowledge Graph; knowledge assertions; provenance links; validity and confidence metadata; retrieval indexes

CMP-09 - Memory System

Memory Graph; memory associations; retention and access metadata; Conversation Objects where assigned

CMP-10 - Learning and Wisdom System

Experience Ledger; Pattern Records; Wisdom Records; learning-run records; promotion decisions

CMP-11 - Temporal Service

Time Facts; Logical clocks; lease records; time-source assessments; Temporal Commitment Records; scheduling constraints; deadline and freshness policies

CMP-12 - Presence and Embodiment Layer

Presence Objects; Embodiment Bindings; Audience Context; Presence Handoffs; modality availability

CMP-13 - Capability Registry

Capability Contracts; Provider Records; Adapter Records; qualification and health metadata

CMP-14 - Action and Transaction Engine

Action Records; Transaction Records; Commit Barrier decisions; idempotency records; rollback and compensation state

CMP-15 - Verification Service

Verification Plans; Verification Results; claim-evaluation state; fidelity and confidence records

CMP-16 - Simulation and Digital Twin Platform

Simulation Runs; Digital Twin State; fidelity models; scenario and isolation records

CMP-17 - Trust Service

Trust Assessments; trust dimensions; confidence and expiry metadata; trust-policy inputs

CMP-18 - Evidence Service

Evidence Objects; custody chains; admission decisions; signature and verification state

CMP-19 - Privacy and Data Governance Service

Data Classification Records; purpose bindings; retention classes; consent and authority records; deletion obligations

CMP-20 - Constitutional Firewall

firewall decisions; cross-domain session state; Treaty enforcement cache; exchange audit bindings; cross-domain incident records

CMP-21 - Treaty Manager

Treaty Records; proposal versions; Owner Authorization Ceremony bindings; activation, suspension, and revocation state

CMP-22 - Event and Messaging Platform

Event Journal; message envelopes; deduplication records; delivery and subscription state; Thread Delivery Records; Transactional Outbox publication state

CMP-23 - State and Persistence Platform

storage engines; commit logs; snapshots; replication metadata; projection and migration state

CMP-24 - Observability and Audit Platform

Audit Ledger; Audit Records; telemetry pipelines; metric and trace stores; alert state

CMP-25 - Runtime Supervisor

Current Runtime Mode; Service Instance Records; Desired and Observed State; health and readiness state; resource assignments; quarantine state

CMP-26 - Recovery Coordinator

Recovery Cases; recovery plans; reconciliation decisions; restoration progress; RPO and RTO evidence

CMP-27 - Human Interaction Layer

Interaction Sessions; turn and modality state; clarification and confirmation records; accessibility preferences

CMP-28 - Self-Model and Constitutional Mirror

Self-Model Records; Constitutional Mirror versions; capability and limitation projections; conformance summaries

CMP-29 - Outcome and Success Evaluation Service

Outcome Objects; success evaluations; side-effect records; review and learning referrals

Appendix B - Interface Handoff to Book IX

305 logical interfaces define semantic outcomes. Book IX owns transport, encoding, field schemas, generated bindings, protocol errors, timeout, retry, and wire compatibility.

Appendix C - Conformance Model

290 conformance tests cover ownership, authority, lifecycle, invariants, failure, recovery, compatibility, privacy, security, and topology. Certification is evidence-based, scoped, attributable, suspendable, and revocable.

Certification status

Final v1.0. 29 components, 348 requirements, 305 logical interfaces, 136 authoritative state domains, and 290 conformance tests. No open Owner Review item.